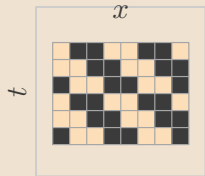


Relative-periodic domain selection by Bernoulli NML

1 Observed spacetime

Choose candidate period and shift.

Example: $p = 2, s = 1$.



Observed CA spacetime U to be explained by a relative-periodic scaffold.

2 Orbit classes

Translate by $(t, x) \mapsto (t + p, x + s)$.

One orbit = one template bit.



The orbit partition turns a spacetime block into a smaller background template.

3 Majority-vote fit

Count ones and zeros inside each orbit.
Majority vote sets the bit.

orbit	count	vote
1	5 / 1	→ ☒
2	1 / 5	→ ☐
3	4 / 2	→ ☒
4	2 / 4	→ ☐

Voted bits form $B^*(p, s)$; the residual mask is $M = U \oplus B^*$.

4 Score and select

Score each candidate with Bernoulli NML.
Lower total is better.

$NML(p, s) = \text{fit} + \text{penalty}$

Fit pure orbits → low entropy

Penalty more template bits → higher cost

Keep the candidate with the smallest total NML.

Output: selected scaffold + residual mask.